

Video coding — techniques, standards and applications

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Virtually all teleconferencing systems incorporating video use compression algorithms to reduce the video data rate to manageable proportions. For a given bit rate, higher compression brings higher quality, although there can also be disadvantages, such as increased delay. Conferencing implies interconnectability, so the development of appropriate standards has played a vital part in enabling the videoconferencing market in particular to become established. This paper describes the main techniques used in video compression and how these have been embodied in successive generations of standards appropriate to various application areas. Other non-standardised techniques, and forthcoming standards are also discussed.

1. Introduction

Teleconferencing has developed through the stages of audio, and video and now data conferencing (including fax, file exchange, shared document editing, etc), and is poised to move on to virtual reality (VR) methods. VR may involve the mixing of real and synthetic video (for example placing the real video images of peoples' head and shoulders round a synthetic conference table), or even just purely synthetic video (for example using completely synthetic representations of people — 'avatars').

In the near future at least, a vital component for teleconferencing will continue to be the ability to *compress* the real video component, which commonly utilises by far the majority of the bandwidth. This paper reviews the fundamentals of video compression, the various standards that have emerged to cater for different application areas, and what the future might hold.

2. Video compression techniques

Analogue video connections for broadcast television consume in the region of 5 MHz bandwidth, but for teleconferencing purposes which require switching and long distance transmission, digital video is the only feasible approach. For digital operation, straightforward pulse code modulation of studio-quality video according to the sampling rates and quantisation law specified in ITU-R Recommendation BT.601 [1] requires a bit rate of 216 Mbit/s.

Clearly, when the above rates are compared to the 64 kbit/s of telephony circuits, the likely tariffs would stifle all but the most enthusiastic and wealthy customer's interest in videoconferencing and videotelephony services! Fortun-

ately, once in a digital form video signals are amenable to compression.

Virtually all the compression algorithms in use today are waveform coders. These attempt to re-create, at the decoder, a replica of the original image waveform by using only the properties of a low-level representation. For example, an image is sampled to produce a set of picture elements, termed pixels, the brightness and colour of each being represented by numbers. Compression is achieved by exploiting properties both of these numbers and of the human visual system. No higher level understanding is undertaken.

An analogy is the difference between text transmitted by a facsimile machine and text transmitted as ASCII characters. The latter is at a much higher level and hence is a more efficient representation. Today, image compression is still the equivalent of the facsimile representation.

If the numbers are reproduced exactly, the coding scheme is said to be lossless and in this case is operating purely on statistical redundancy. Techniques which remove visual redundancy reconstruct images which differ objectively from the originals and are classified as lossy. The errors are either subjectively invisible or at an acceptably low level. There is no rigid definition of how much distortion is permissible because the acceptability is subjective, depending on many factors, such as the picture material, the viewing conditions, the viewer's eyesight, the use to which the pictures are being put and, not least, the cost.

2.1 Removal of statistical redundancy

Statistical redundancy is not unique to image data processing. Many computer users, on finding that the capacity of their hard disk has become insufficient, have installed software which allows more data to be stored, or used file compression utilities to send files more efficiently by e-mail. Similarly most PSTN modems now incorporate data compression [2] which increases the apparent transmission speed.

Many of these techniques are based on the Ziv-Lempel algorithm [3] which recognises repetitions of parts of the data and sends a pointer to earlier occurrences of that data, rather than repeating the data in full. Depending on the characteristics of an image (natural or synthetic, background noise level, etc), the Ziv-Lempel algorithm may provide compression in the range just above unity up to three times.

However, the above algorithm does not make effective use of the fact that image pixels tend to be highly correlated with their neighbours in space or time, which is the basis of many compression techniques that have been developed specifically for images and data from other natural sources (for example, audio).

Predictive coding

The basis of predictive coding is to use pixels or data already received to form an estimate or 'prediction' of the next pixels or data to be transmitted. A basic form of predictive coding is illustrated in Fig 1. Pixels are fed into the system in raster scanned order (i.e pixel by pixel scanned left to right across the image, line by line down the image, and picture by picture in a moving sequence). Differences are formed between each pixel and a nearby one temporarily stored in the delay element. The delay element is fixed and corresponds to selecting the pixel immediately to the left, the pixel immediately above (both intraframe predictions) or, in motion video, the pixel in the same place in the previous picture (interframe prediction).

The difference or 'prediction error' is seldom exactly zero because even in regions with no detail or in stationary areas the camera system usually adds some noise. However, the variance of the prediction error is generally substantially less than that of the original and further techniques such as entropy coding (see later in this section) can be used to represent the data with fewer bits than the original pixels. The original pixels can be reconstructed from the prediction errors by a decoder which consists of just the components in the outlined box.

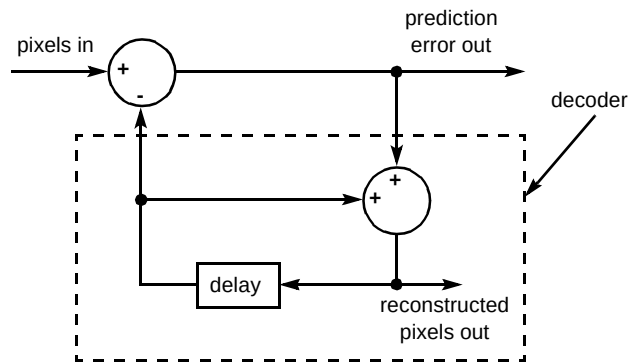


Fig 1 Basic predictive coder.

As well as using just one previous pixel as the predictor, more complex schemes employing several previous pixels together may be used, for example a local average of a few spatially adjacent pixels may provide a better estimate of the new one by averaging out noise, or estimating and extrapolating the local slope of the image function.

When predicting from one picture to the next in moving video, the single pixel predictor is of course very good in regions of the image where there is no change (or 'motion'), but often very bad where there is actually motion. Unfortunately the latter is precisely the area where most data needs to be sent and compression is most needed! This observation led to the introduction of conditional replenishment, whereby in areas of the picture where there is no temporal change at all, it is sufficient for a decoder to simply display the pixels from the previous frame again. In changed areas, new pixels are transmitted using prediction from spatially adjacent pixels. Omitting the redundant transmission of unchanged pixels gives a big gain in compression.

To achieve higher compression in the changed areas, motion compensated prediction was developed.

Motion compensation

This technique is an extension of the above prediction method, in which the delay for each pixel is varied dynamically throughout each still image to find the place in the previous still which is the best prediction. The corresponding horizontal and vertical offsets to refer to that best prediction are included in the coded bit stream. Ideally these offsets would be sent for every pixel but the overhead of doing this is much more than the savings obtained from the smaller prediction error. Instead, a single pair of offsets is applied to many pixels, a 16×16 block of them being a common compromise.

Motion-compensated prediction works fairly well on videoconference type scenes where the gross movements of people's heads and bodies are represented to a first

approximation by simple translations of their positions from the previous image. However, real world objects also undergo rotation, deformation, occlusion and so on, which are not well modelled by the above block-based motion compensation. More complex schemes have been devised to model these higher order changes in image structure, including representing the motion by complex warping functions, and delineating the boundaries between areas with different motions.

Entropy coding

Entropy coding is a general term for lossless data compression methods which rely on the statistics of a set of 'events' to be compressed. In most practical cases this means the overall frequency of occurrence of the various events in a set. The length of the codeword used to convey a particular event is matched to the likelihood of it occurring. Shorter codes are used for the frequent events and longer codes for those appearing less often, hence the term 'variable length coding'. For example, in Morse code a single dot represents the frequently occurring letter 'e' whereas the rarer letter 'q' is encoded as dash, dash, dot, dash.

It can be shown [4] that the optimum length of a codeword for an event of probability p is $\log_2(1/p)$. If all possible events in the set are assigned codewords according to this formula, the overall bit rate required to send events from that set with the given probabilities will be minimised. Thus for an event with a probability of occurrence of $1/8$ the optimum codeword length is 3 bits. Unfortunately in practical cases the formula will yield non-integer values for codeword lengths, and a means is needed to optimally assign integer length codewords to events. Huffman [5] devised an algorithm to do this, hence the term Huffman coding (which is often wrongly used as a term for variable length coding, which need not, in general, be optimal).

Huffman's method assigns integer length codewords to events, but this results in a loss of efficiency since the theoretically optimal codeword lengths are non-integer. Arithmetic coding [6] is a technique which overcomes this by not having a one to one mapping of events to codewords, and thus comes closer to the optimum compression. However, it is also more difficult to resynchronise the decoder in the presence of errors.

Entropy coding gives very little compression if applied directly to image signals because the distribution of the brightness levels is fairly uniform. However, prediction errors have a very peaked distribution centred about zero and variable length coding is very worthwhile.

2.2 Removal of perceptual redundancy

The degree of compression obtainable from lossless techniques is modest, typically rather less than 5:1. To

achieve lower bit rates, it is necessary to employ lossy compression methods. Most complete algorithms incorporate both types.

Subsampling

A straightforward approach is to reduce the number of samples in the originals. After decoding, the missing samples can be replaced by repeating, or, better still, interpolating from, those that were coded. This lowers the spatial or temporal resolution but may not materially affect the usability of the resulting pictures. Unfortunately, the coded bit rate does not decrease linearly with the sample rate because the fewer samples have less correlation and are less amenable to compression.

Colour images are usually represented by three colour components. Red, green and blue are used for image capture and display purposes, but for transmission and compression an alternative format known as YUV is preferred. RGB can be converted to YUV and back by a linear matrix operation. The Y or luminance component represents the brightness, whereas U and V are 'colour difference' signals, which both have zero value in black, grey and white parts of the image, and are non-zero where colour is present¹. Because the human eye has a higher visual acuity for the luminance component of images compared to the colour component it is possible to reduce the spatial resolution of the U and V signals significantly compared to the Y component with no perceptible loss of quality. In videoconference applications this has led to the use of image formats such as common intermediate format (CIF) where the Y resolution is 352 by 288 pixels, and U and V are reduced to 176 by 144 pixels each.

Quantisation

Quantisation reduces the number of discrete values a variable can take, reducing the number of bits needed to encode it. Accuracy is lost — a number of different input values will be the same at the output. Recommendation BT.601 allocates 8 bits each to the Y, U and V components, this being near the minimum necessary to avoid a 'painting by numbers' effect called contouring (although greater numbers of bits are often used during editing and manipulation stages to ensure adequate resolution is retained in the final result). There is not much scope for further quantising the original pictures directly without introducing visible degradation.

This barrier does not apply to prediction errors, which are generally largest at edges or fine detail in the originals. The human visual system is much less sensitive to errors in high detail regions — a phenomenon called spatial masking. This can be exploited by applying a nonlinear quantiser to

¹ To be more precise, Y, U and V are usually represented as 8-bit numbers, which are interpreted as the range 0 to 255. U and V take the mid-point value of 128 in black/grey/white regions, and fluctuate above and below 128 in colour regions, so 128 is their notional 'zero' value.

the prediction error so that large errors are encoded less accurately than small ones.

Transform coding

Transform coding is a mathematical operation which transforms a set of numbers to another set with more favourable properties, while retaining the same information. The discrete cosine transform (DCT) is one of the more popular transforms for image compression.

The DCT is usually applied to a block, typically 8×8 , of original image pixels or prediction errors, and produces an equal number of coefficients which are in many ways similar to Fourier transform coefficients. They contain, in order, increasingly higher frequency components of the original data.

For natural images, these DCT coefficients have the useful property that over a large ensemble of blocks, the probability distributions of the coefficient values are highly peaked around the value zero, and can be efficiently compressed using entropy coding. Additionally, the variance of the probability distributions decreases with increasing spatial frequency of the coefficients such that after quantisation, many of the coefficients are zero and can be discarded. This is illustrated in Fig 2, where a complex dark to light shading across an 8×8 pixel block results in only a few DCT coefficients having significant non-zero values. Furthermore, the eye is less sensitive to the higher spatial frequencies allowing them to be more coarsely quantised, resulting in additional compression.

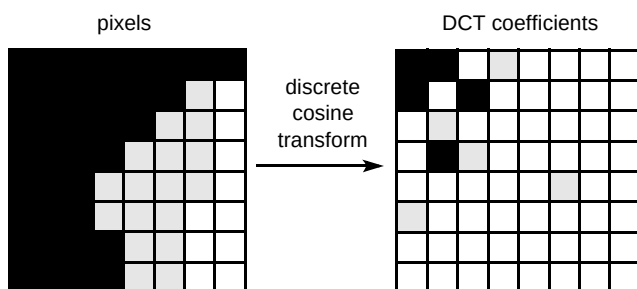


Fig 2 Discrete cosine transform of an 8×8 block of pixels.

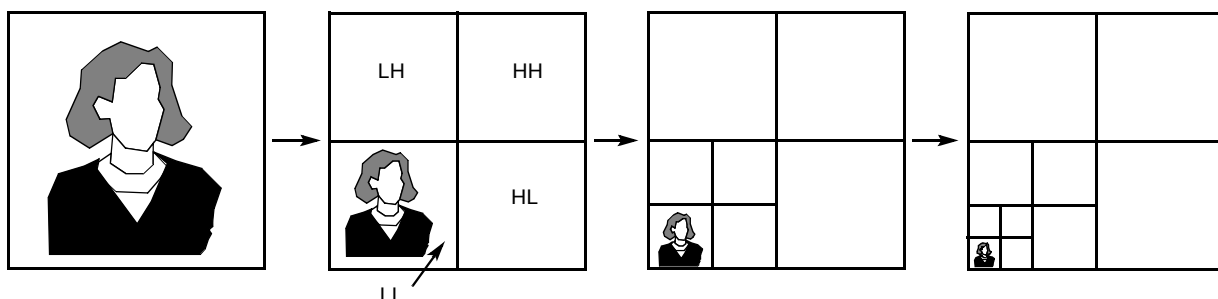


Fig 3 Progressive stages of wavelet transform of an image.

Wavelets

Wavelets [7] are another way of transforming an image into an alternative representation based on spatial frequencies, in a similar manner to the DCT. However, whereas the DCT is performed on a block of pixels at a time to produce a block of DCT coefficients representing the spatial frequencies within the block, wavelets are usually used on the complete image to split it into two spatial frequency bands in each direction (usually known as the L or low and H or high bands), yielding a total of four sub-bands.

These four bands are designated LL, LH, HL, and HH. The LL band is essentially the original image reduced in scale by a factor of two in both directions, whereas the other bands represent higher frequency details associated with horizontal, vertical and diagonal edges or other detail. The LL band is usually subject to one or more further rounds of wavelet decomposition, resulting in a hierarchy of bands as illustrated in Fig 3.

As with the DCT, the wavelet transform itself gives no compression directly, but it maps the image into a domain where the significant data is structured in a way which is amenable to compression. Thus, apart from the LL band, the resultant wavelet coefficients have highly peaked distributions around zero (as with the DCT coefficients), and entropy coding can be used. Secondly, the locations of non-zero coefficients in the various bands are usually spatially aligned and quadtree type addressing schemes can efficiently indicate those coefficients to be transmitted with low overhead [8]. Thirdly, the hierarchy of spatial frequencies provided by the wavelet transform are a good match on to the spatial frequency sensitivity of the human eye, allowing coarser quantisation to be used on the higher frequencies.

Because the wavelet transform operates equally on all pixels in the image rather than grouping them arbitrarily into blocks, the coding distortions produced by excessive quantisation tend to be less visible and objectionable than those produced by block transform coders, where the block structure itself often becomes visible at high compression.

Vector quantisation

Wavelets and the DCT exploit the correlation of adjacent pixels by generating coefficients that have small or zero value if adjacent pixels are similar. Another way to exploit this correlation is vector quantisation (VQ) [9]. A block of pixels or prediction errors is quantised as one unit (a vector), rather than a pixel at a time. Typically, 4×4 sized blocks might be used, and a codebook of hundreds or thousands of possible example block patterns would be searched to find the pattern giving the best match. Only the index number of that entry in the codebook need then be transmitted.

VQ has a simple decode operation, being merely a table look-up, but the encoding process can be very demanding because at worst each block will require a full search of the codebook. In practice, various ways of pre-normalising the data, and structuring the codebooks can reduce this problem. VQ can also be used to quantise the data resulting from other processes such as DCT or wavelet transformation, and both theoretically and practically should perform better than an equivalent number of independent scalar quantisers. However, there are some problems concerning construction of codebooks to adequately cover the range of vectors to be quantised, and of the processing power required to perform the codebook searching at the encoder.

Fractal image coding

Fractal coding is another way of exploiting redundancy in images. It is similar to VQ except that no explicit codebook is required. It relies on the fact that in any given image it is usually possible to find one part of the image which, when suitably scaled down in size, rotated and grey-scale adjusted, provides a very close approximation to another part [10, 11]. By dividing an image into small regions (usually square blocks) and finding the appropriate 'mapping function' for each region, the whole image can be represented purely as a set of mapping functions. The surprising thing is that as long as these mapping functions are contractive (meaning essentially that the size and grey level must always be scaled down in the mapping) no actual image data at all is required to initialise the process. By starting with any image and applying the mapping functions repeatedly, the resulting image will finally converge to one close to the original image used to select the mappings. Depending on the scalings this would typically take less than ten iterations of each region mapping to converge to the endpoint.

Fractal coding can achieve very high compression in specially selected cases, but for typical images it is comparable to other methods. However, it has some other useful properties:

- it retains edge sharpness well,
- on natural images it can generate textures that although not faithful to the original image are quite often subjectively acceptable,
- the mappings found during compression can be used to scale images up or down in pixel resolution — although scaling up cannot accurately regenerate information previously discarded, the results can be subjectively pleasing,
- the decoding process is simple.

A major drawback is that the encoding process is very computationally intensive and to achieve real-time encoding it is currently necessary to compromise the compression efficiency by simplifying the process.

3. Standards and applications

The preceding section has highlighted most of the commonly used image compression techniques. This section reviews the historical development of image compression standards.

After some proprietary videoconferencing codecs operating at 6 Mbit/s from US and Japan, the European COST 211 project developed by 1984 a 2 Mbit/s codec [12] which was subsequently adopted by the CCITT as Recommendation H.120. This was based on conditional replenishment, changed areas being updated using an intraframe pixel prediction method.

A few years later more sophisticated techniques were applied in the development of ITU-T Recommendation H.261 [13]. This provides the motion video component of ITU-T Recommendation H.320 [14] for videophone and videoconferencing services at total bit rates between 64 and 1920 kbit/s, and embodies the basic principles from which subsequent algorithms including MPEG 1 and 2, and H.263 were developed through a process of progressive refinement and addition of new features.

3.1 H.261

The picture format used for H.261 is either CIF (see section 2.2) or Quarter CIF (QCIF) where the numbers of pixels in each dimension are half those of CIF. Remembering that the colour components U and V are at half the spatial resolution of the Y pixels, Y, U and V pixels are grouped separately into 8×8 blocks, and then four Y blocks and the spatially corresponding U and V blocks are associated into a 'macroblock' as shown in Fig 4.

The structure of an H.261 encoder is shown in Fig 5 and can be seen to combine several of the techniques discussed in section 2.

The encoder works as a predictive coding loop by analogy with Fig 1. It uses past data to form a prediction of

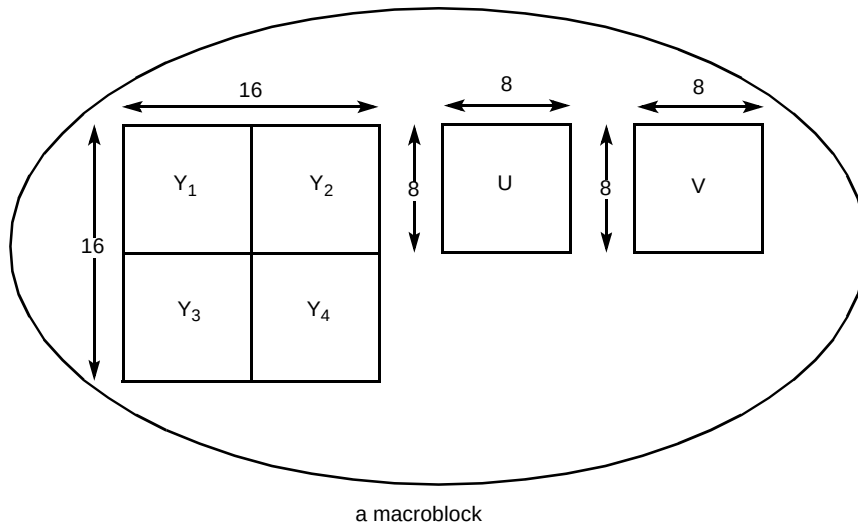


Fig 4 A macroblock composed of four Y blocks and a U and a V block

the current input data and quantises the prediction error for transmission. It then performs the 'decoder' functions of inverse quantisation and summation with the prediction to reconstruct an approximation to the input data.

To understand the coding loop in more detail, begin with the picture store, which holds the previously decoded picture. The motion estimation unit compares the incoming picture, a macroblock at a time, with this stored picture to determine the best motion vector to use for each macroblock. Having done so, it directs the picture store to output pixels (an 8×8 block at a time) to the rest of the coding loop, using the previously calculated motion vectors to select the appropriate parts of the stored picture. When the switch is in the 'inter' position, a block passes through a

low-pass spatial filter which has been found to improve coding performance, and is then used as the prediction for the corresponding input block. After subtraction to yield the prediction error, the block undergoes a discrete cosine transform, the DCT coefficients are quantised, and the non-zero coefficients addressed and passed to the variable length coder. An inverse quantiser generates the appropriate reconstruction levels for the quantised coefficients and passes them through the inverse DCT to give an approximation to the prediction error, with some added noise due to the quantisation process. On adding this to the prediction, the reconstructed picture is produced, which is close to the input picture, but with the added quantisation noise. As this noise is generated in the transform domain, it appears in the image as a fairly evenly distributed random component,

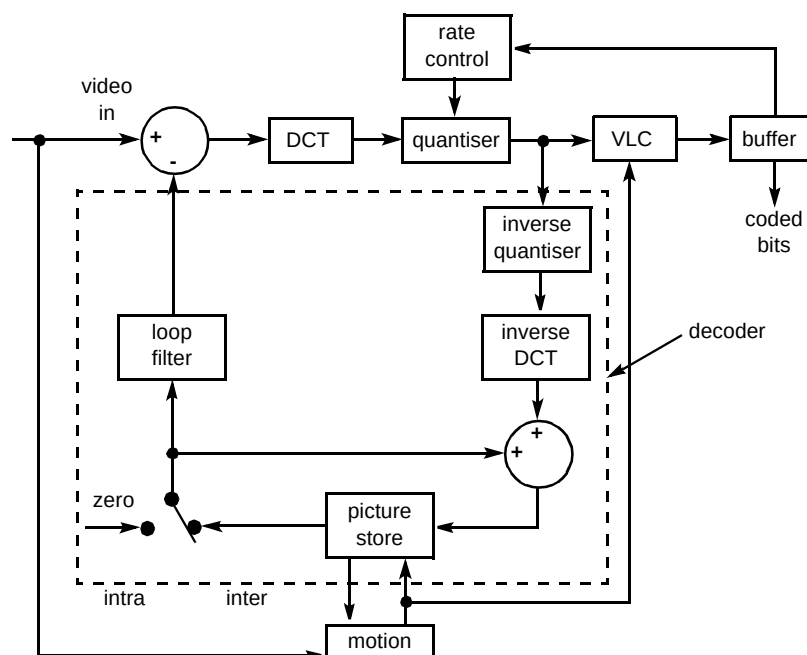


Fig 5 H.261 encoder.

though, at high compression, blocking and ringing artefacts are also visible.

The encoder is completed by a buffer which receives data generated by the coding loop at a variable rate and outputs it at a usually constant channel rate, and a rate control mechanism which adjusts the quantiser and also usually the coded picture rate to keep the long-term output bit rate the same as the channel rate, and the short-term variation within the range of variation that the buffer can absorb.

The H.261 decoder consists of the components within the outlined box in Fig 5, preceded by an input buffer and variable length decoder. As the H.261 decoder does not have to perform the motion estimation function or the forward DCT, it requires much less processing power than the encoder.

The prediction generated within an encoder must match that generated by a decoder, or else the contents of the two respective frame stores will progressively diverge as more frames are coded. Ideally the 'decoder' part of the encoder loop would be defined to be mathematically exactly equivalent to all decoders. However, in practice there are several ways of implementing the DCT which, because of the effects of rounding errors, can result in small differences in the numbers produced in different systems. Although the magnitude of this error is specified and constrained by the H.261 specification of the DCT accuracy, it can still accumulate and cause encoder/decoder mismatch. As a means of constraining this divergence, in addition to the predictive- or inter-coding mode, the encoder can also use the switch in Fig 5 to set the prediction to zero for a macroblock or a complete picture, in which case the input signal is coded directly in 'intra' mode. Since this mode does not rely on previously transmitted data it is used to initialise the encoder and decoder loops to the same state at the start of a session, to clear any transmission errors that may have resulted in the encoder and decoder loops getting out of step, and to control long-term build-up of rounding errors in the DCT. Intra mode is not used more often than necessary as it does not provide as much compression as inter mode.

The H.261 coder is only formally specified to work on CIF and QCIF pictures, and at bit rates in the range 64 kbit/s

up to 1920 kbit/s. Although it can be used outside this range, other codecs such as MPEG 1 and 2 are generally more suitable at higher rates, and H.263 at lower rates. The main application of H.261 is still videoconferencing using the H.320 Recommendation. On a 2B channel ISDN connection the speech is usually coded with G.728 at 16 kbit/s leaving 108.8 kbit/s for the video (after deducting multiplexing overheads), at which rate H.261 provides adequate quality to cope with for two or three people at each terminal with moderate amounts of motion. For higher numbers of people per terminal, or for longer conferences, a higher bit rate provides higher quality and less user strain. H.261 is now also being used in other areas including streaming of video from web sites, and videoconferencing using local area networks (H.323).

3.2 MPEG-1

The H.261 coding methods were further refined in ISO MPEG-1 [15] and MPEG-2 [16], which are aimed at a wider range of applications including those where storage is an important aspect.

MPEG-1 added the technique of half-pixel motion estimation to H.261, which used only integer pixel estimation. This allowed motions to be more finely tracked and improved the prediction and hence compression performance. The second main change was to add the concept of 'B-Pictures'. These are pictures in which predictions can be formed from a picture occurring later in time than the current picture, instead of just from the previous picture as was the case with H.261. Using such bi-directional predictions (hence B-Pictures) improves the overall performance. The ability to look ahead in time is accomplished by storing up a short set of pictures in frame buffers, coding the latest stored picture in the normal forward prediction (P-Picture) mode, then going back and coding the remaining stored ones in B-picture mode (Fig 6). Although B-Pictures offer significantly higher compression than P-Pictures (by a factor of typically 2 to 3), which in turn have higher compression than I -Pictures (by a similar factor), there is a penalty of increased picture delay, and more complex processing requirements. The delay of several pictures (equivalent to typically 100 ms) is usually only a problem for real time conversational services such as tele-conferencing, for which the delay can be minimised at the expense of compression by using few or no B-Pictures.

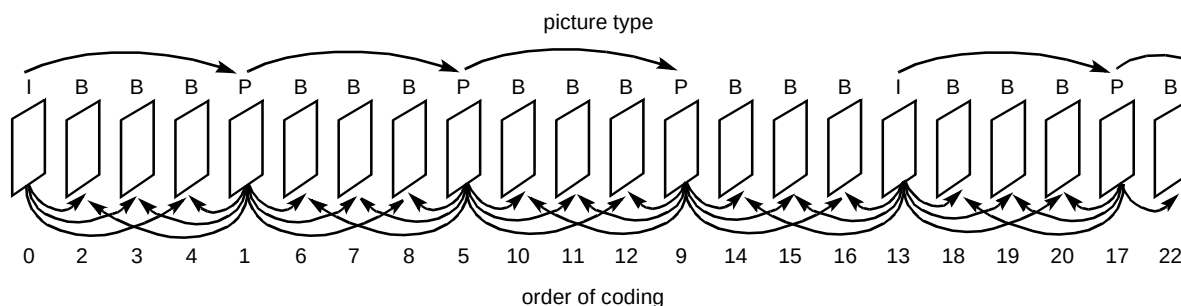


Fig 6 Order of coding and prediction paths of I, P and B-pictures.

The compression achieved by MPEG-1 is around 30% greater than that of H.261, though this is somewhat dependent on picture material and bit rate.

Figure 6 also illustrates the regular insertion of intra-coded pictures (I-Pictures) in a typical MPEG sequence. Since these can be decoded independently of any preceding or following frames they are used to ensure rapid recovery from any transmissions errors, and allow random access (i.e. jumping into the compressed video stream at any point). They also provide a means of playing the video sequence backwards as is necessary to duplicate video cassette recorder functionality such as fast reverse play — essential for video-on-demand type applications and for general browsing of compressed video. MPEG-1 was originally developed for use at single-speed CD ROM bit rates of around 1.2 Mbit/s and CIF resolution. It is, however, suitable for other applications not requiring interlaced pictures (see section 3.3), such as the interactive multimedia services trials run by BT [17] where a video bit rate of 1.6 Mbit/s was used to deliver roughly VHS quality video to homes for a mix of entertainment, education and information.

3.3 MPEG-2

MPEG-2 was developed in collaboration with ITU-T and is also known as ITU-T Recommendation H.262. It extended MPEG-1 to deal with the interlaced scanning of conventional television systems. Interlace is a hangover from analogue transmission and display systems which addressed the problem that for a good rendition of motion only 25 to 30 pictures per second are required, whereas at least 50 screen refreshes per second are required to avoid annoying flickering. In cinema this is done by displaying each picture several times. In analogue television there was no means of storing a picture at the receiver for repeated display, so each picture or 'frame' was scanned as two 'fields', each using half the number of lines in the frame, with the lines spatially interlaced to give double the number of lines for the frame, effectively increasing the vertical resolution. So 50 (or, in the USA-developed NTSC system, 60) fields per second are delivered to reduce screen flicker, at 25 (or 30) frames per second. When digitally coding such pictures it is important to realise that the two fields were captured at different times, and thus any moving object appears at different positions in them. If the fields were coded as a single frame there would be some very complex high-spatial frequencies to deal with as the alternate lines of the fields contain uncorrelated information. MPEG-2 deals with this by having extra prediction modes which can predict from one field to another in appropriate ways to extract the useful correlations remaining. By coding interlaced TV signals, MPEG-2 can be used to deliver broadcast (BT.601) resolution, at around 4 to 8 Mbit/s. MPEG-2 can also be used to encode higher resolutions such as HDTV, which was to be the subject of a separate 'MPEG-3' phase, but was subsumed into MPEG-2.

As with all video coding schemes, the quality achieved depends on the complexity and motion in the pictures, and on the available bit rate. Since historically the networks available have operated at fixed bit rates it has been usual to operate video compression schemes with a buffer and control system to meet that fixed bit rate. However, the result is that the video quality actually varies through the sequence — dropping in the more complex parts and perhaps using more bits than is necessary for acceptable quality in others.

With the arrival of networks that can support variable bit rate transmission such as ATM, it should be possible to operate with constant picture quality by allowing the bit rate to vary as necessary. Similar ideas can be used on satellite or cable delivery systems where a total bit rate budget can be shared adaptively between several video channels according to their relative need, achieving a higher overall quality than a fixed bit rate budget for each. Thus in reality it is difficult to put a single figure on the bit rate required for 'broadcast quality video', and the 4 Mbit/s mentioned above would be adequate for most purposes, but will not prevent some visible impairments on complex scenes such as sports with rapid camera and player motions.

Another new feature introduced by MPEG-2 is 'scalability', whereby one video bit stream can be used to bootstrap the quality of pictures provided by a second video bit stream, and so on. An alternative view is that one compressed stream contains within itself sub-streams which can also be decoded to give lower quality pictures than the full stream. This scalability involves using the decoded pictures of the lower bit rate stream as sources of prediction for the next layer, which can be configured to provide a higher spatial resolution, a higher temporal resolution (more pictures/second), higher quality (less distortions), or a combination of these. A drawback is that the quality achieved by using all the layers is usually lower than that achieved in a single layered encoding at the same bit rate.

There are several applications for scalability. One is where a network may offer several levels of service such that a lower bit rate stream is guaranteed timely arrival, whereas one or more higher rate streams may suffer some data loss. The lowest layer guarantees that a picture is always available for display, while the other layers enhance the quality in a way that is resilient to occasional data loss. This scenario can apply in ATM and in Internet protocol (IP) networks.

A second application is multicasting on IP networks, where a single server sends out the multiple layers, and clients receive as many layers as network conditions allow them to, adding and dropping layers dynamically as network congestion changes, to receive the best quality video possible at any one time.

3.4 H.263

With the advent of faster modems, especially V.34 yielding 28.8 kbit/s [18], V.34 extensions to 33.6 kbit/s (full duplex) and 56 kbit/s (though not bi-directional at this rate), the possibility of providing useful motion video over the PSTN has emerged. The experience gained from H.261 and MPEG has been further augmented by development of the H.263 video compression algorithm [19]. The main aim for this algorithm was higher compression, given the limited bit rate available on the PSTN, yet low delay as the main application was conversational services. This aim was largely achieved as it delivers around twice the compression of H.261, though, as with the gain provided by MPEG-1, this is dependent on picture material and bit rate.

A significant departure for H.263 was to make some of the coding enhancements optional, to be negotiated between an encoder and a decoder. These optional modes appear as annexes to H.263.

- Annex D — unrestricted motion vector mode

This allows motion vectors to point outside the normal picture boundaries by extrapolating out the edge pixels, and is particularly useful if the camera or whole scene is moving. This mode also doubles the allowed range of motion vectors from 16 to 32 pixels.

- Annex E — syntax-based arithmetic coding mode

This substitutes arithmetic coding for the usual variable length coding procedure, and achieves slightly higher compression efficiency.

- Annex F — advanced prediction mode

This allows motion compensation on 8 by 8 blocks as well as the usual 16 by 16 ones, and also uses a technique called overlapped block motion compensation, whereby a degree of overlap of adjacent compensated blocks is used to smooth out some of the block edges otherwise generated by block-based motion compensation.

- Annex G — PB-frames mode

This allows the use of bi-directional prediction (see section 3.2) for increased compression. In contrast to the MPEG B-Pictures, here a P and a B picture are combined together into one data structure.

Figures for the amount of extra compression achieved by and the subjective effect of each Annex on different types of sequence are given in Whybray and Ellis [20]. Figure 7 shows a comparison between H.261, the base level of H.263, and H.263 with all four optional annexes (as described above) turned on.

The intended application for H.263 was the H.324 PSTN videophone Recommendation, which was ratified by the ITU in 1996, with products appearing in early 1997. Many of these H.324-compliant videophones were implemented as PC applications, requiring only a suitable modem and camera interface card to convert a multimedia PC into a videophone, with the video coding algorithm implemented in software on the PC. H.263 has also found favour with organisations providing streamed video on the Internet, where again the high compression efficiency is a benefit.

Further enhancements to H.263 (the main ones being listed below) are under development in the ITU in draft form at the time of writing, and are due for formal ratification (Decision, in ITU terms) in January 1998.

- New picture types:

- scalability — use of enhancement layers of H.263 coded video to build higher resolutions/qualities at high bit rates in an embedded manner,
- custom source format — enables coding of pictures with arbitrary frame sizes, and arbitrary pixel aspect ratios,
- improved PB frames — improved version of an existing picture type for higher compression.

- New coding modes:

- advanced intra coding — improved compression of intra frames,
- deblocking filter — reduces blocking artefacts to improve subjective quality,
- slice structure — groups macroblocks for improved error resilience,
- reference picture selection — speeds up recovery from channel errors by use of a back channel,
- reference picture resampling — improves prediction process by allowing affine transformation of prediction picture,
- reduced resolution update — helps maintain a high frame rate where there is high motion,
- independent segment decoding — improves error resilience,
- alternate inter VLC — slightly improves inter compression,
- modified quantisation — allows more rapid control of the quantiser, and increases quantiser range.



Fig 7 Stills from sequences coded at 28.8 kbit/s, 12.5 pictures/second. The four pictures are: (a) original frame (b) H.261 (c) H.263 base level and (d) H.263 with options D, E, F and G on [20].

The large number of these optional modes extends the application areas for H.263, but has caused concerns about compatibility. The existing scenario in conversational services is that the end-points should negotiate (normally using ITU Recommendation H.245) which modes are common and can be used, though the complexity of this negotiation may be getting out of hand. For other applications where there is limited scope for negotiation, for example streaming of pre-compressed video from a server, the decoder must be able to handle all modes used by the encoder.

3.5 JPEG

Regarding still pictures, in the 1970s, several telcos had introduced or were considering videotext services. Using similar formats to teletext, these were character based, providing text or block mosaic graphics. There was a desire to add 'photographic quality' pictures but a full screen image at Recommendation BT.601 quality is more than 800 Kbyte. At the then typical modem rate of 1200 baud, such a still would have taken about 100 minutes to receive. Even with smaller images and some relaxation of their quality, the waiting times would have been unacceptable.

With photographic videotext as a catalyst, the usefulness of a generic compression standard for still images was widely appreciated. As a result the ISO and ITU-T jointly developed the JPEG algorithms [21, 22] which are equally suited to both storage and transmission applications. The basic JPEG algorithm is very similar to the intra-frame coding modes of the moving picture standards, being based on an 8×8 DCT. In teleconferencing, JPEG will often be used as the compression method for sending still images between conference participants as part of some higher level teleconferencing application.

4. Video transcoding

Compression algorithms are normally used to compress video to a known target bit rate, either for immediate transmission at that bit rate, or for storage and later playback. Once the compressed bit stream has been produced, it is usually not possible to directly access and decode it at a different bit rate. However, in the future it will increasingly be necessary to do this to support services such as:

- video databases which can be accessed at multiple bit rates,

- interworking adapters between networks carrying video at different rates, for example PSTN/ISDN,
- continuous presence multipoint videoconference units, where the input and output bit rates may differ.

The latter two are particularly important for teleconferencing applications.

In general, changing to a lower rate can only be accomplished by decoding the bit stream back to images and then re-encoding. It was previously assumed that the delays of the two tandemed pairs of coding and decoding operations must be additive and therefore may become unacceptable in some interactive applications. However, investigations by BT have shown that transcoders can be constructed such that the total delay from the input to the first encoder, through the transcoder, to the output of the final decoder is substantially less than the sum of the delays through two equivalent but independent encode-plus-decode pairs [23]. In fact the level of delay can decrease to that of the original encoder plus decoder.

The key lies in recognising that the vast majority of the unavoidable delay, that which cannot be lessened by more powerful hardware, in a coder plus decoder combination, arises from the rate smoothing buffers. These are needed to match the varying rates of bit generation by the coding kernel and bit consumption by the decoding kernel to constant rate transmission channels. These kernels and buffers are illustrated in Fig 8. In Fig 9 the buffers in the transcoder are rearranged to make them contiguous, by operating on uncompressed instead of compressed video. By precisely controlling the coding kernel in the transcoder

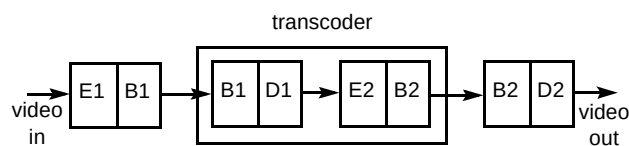


Fig 8 Encoder (E) and decoder (D) kernels and their buffers (B).

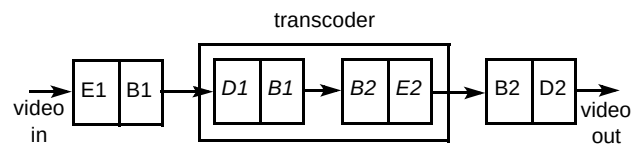


Fig 9 Encoder (E) and decoder (D) with adjacent buffers operating on uncompressed video data.

so that the fill levels of the two buffers are complementary, the total delay through them becomes constant and they can be removed.

In an extension of this technique the coding kernel in the transcoder is greatly simplified by using the motion vectors from the originating encoder. The bit rate reduction is achieved by requantising of DCT coefficients. However, this causes problems for predictive schemes because the prediction loops at the original encoder and at the final decoder are operating with different data, allowing the loops to diverge and distortion to build up at the final decoder. BT has devised the technique, shown in Fig 10, of introducing a correction signal at the transcoder. This does not eliminate the error but continuously attempts to reduce its size, thus counteracting its growth. The result is a bit stream which has been further compressed, but with very little extra delay, low complexity compared to a full decode and re-encode, and a quality only slightly lower than if the full

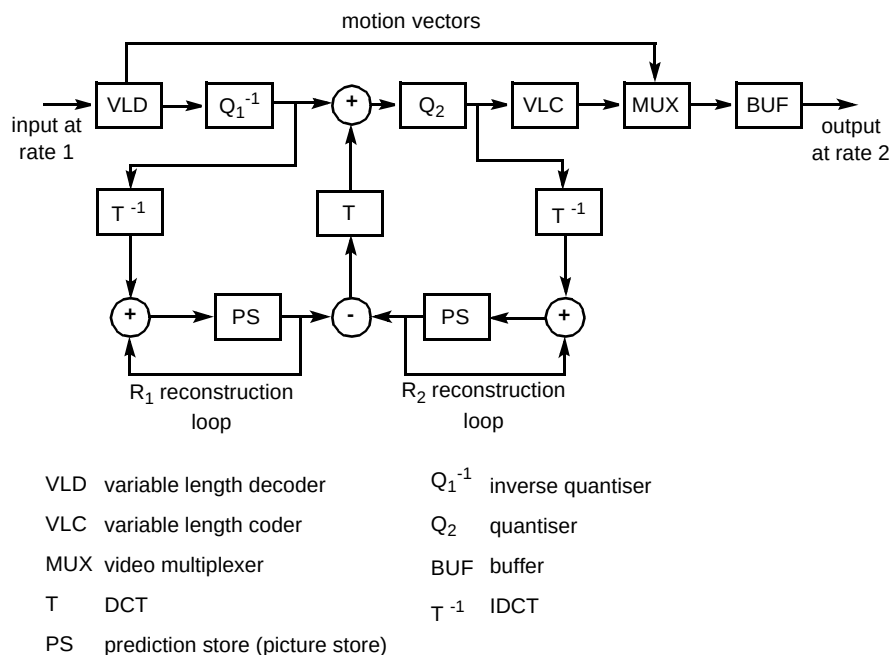


Fig 10 Transcoder for bit rate reduction with drift correction.

compression had been performed in one pass. As the new services mentioned above become more widely used, transcoding will become increasingly commonplace.

5. Future coding algorithms

There is general agreement in the image processing community that existing algorithms are all approaching a limit of compression dictated by purely statistical analysis of images, and exploitation of the human visual system's characteristics. Any large improvements in compression ratios will now require a deeper consideration of the nature of images, in effect using prior knowledge to enable a higher level of abstraction of the image data to be achieved. For example, most images are formed as the projection by a lens of objects in a 3-dimensional world into a 2-dimensional image. There is a direct correspondence between real-world objects and regions in the image. Since the original objects often have some continuity of shade, motion, texture, shape and so on, this continuity is mapped into the resultant image. Conventional coding algorithms such as predictive coding and motion compensation take some advantage of these properties, but do not make full use of the object-like nature of the real world. Admittedly, not everything is 'objects' — water, grass, skin for example are better described as textures, but can usually be segmented into discrete regions in a 2-dimensional image.

Algorithms are being developed which segment each image in a video sequence into regions relating as much as possible to real-world objects, and by analysing the motions of these regions can describe the video sequence in a very compact form [24]. Currently this typically relies on warping the objects in one frame into their new shapes and positions in the next frame, then sending whatever small additional information is necessary to correct any areas not well handled by this method — such as new surfaces revealed by object rotations. It is anticipated that this technique will develop to include full 3-dimensional modelling of the original objects, allowing more accurate reconstruction and higher compression.

Some extreme examples of this form of coding have already been developed in specialised areas. For example, 3-dimensional computer models of human heads have been used to enable synthesis of realistically animated images of human heads talking, moving around naturally, and with facial expressions [25].

Particularly life-like results are achieved if the image of a real person is mapped on to the model. The bit rate required to animate the head is of the order of only a few hundred bits per second. Although synthesis of a human head is now fairly straightforward, analysis of an image of a person's head to extract the motion, mouth shape and facial expression accurately enough to drive the model remains only partly solved.

Current standards work on advanced coding algorithms is centred around MPEG-4.

5.1 MPEG-4

ISO MPEG-4 is an emerging multimedia coding standard that, in addition to higher compression, aims to support new, content-based tools for communication, access and manipulation of digital audiovisual data. As the traditional boundaries between the computer, telecommunications and TV/film industries are blurring, elements that have previously belonged to each of the areas are being introduced into the other two. Also there are major trends towards wireless communications, interactive computer applications and integration of audiovisual data into more and more applications. MPEG-4 is seeking to address the new expectations and requirements emerging from these developments — for example, to exploit the opportunities offered by low-cost, high-performance computing technology and rapid proliferation of multimedia databases.

Early work on requirements for MPEG-4 video identified eight key functionalities which were not thought to be well supported by existing or other emerging standards. These were grouped into the three fundamental categories of 'Content-Based Interactivity', 'Compression Functionality' and 'Universal Access Functionality' (see Table 1).

Table 1 MPEG-4 video categories.

Categories	Functionalities
Content-based interactivity	Content-based multimedia data access tools Content-based manipulation and bit stream editing Hybrid natural and synthetic data coding Improved temporal random access
Compression functionality	Improved coding efficiency Coding of multiple concurrent data streams
Universal access functionality	Robustness in error-prone environments Content-based scalability

The full MPEG-4 standard will comprise six parts, of which systems, audio and visual are the most advanced at the time of writing. Other parts include a conformance testing specification, a technical report on reference software implementations and a multimedia integration framework definition. The systems, audio and visual parts are currently at working draft (WD) status and are expected to progress to full international standard (IS) status in January 1999. Audio (WD 14496-3) and visual (WD 14496-2) will define a standardised coded representation of audio and visual content, both natural and synthetic, called 'audio-visual objects' or AVOs. Systems (WD 14496-1) will

standardise the composition of these objects together to form compound AVOs (e.g. an audiovisual scene), and multiplex and synchronise the data associated with individual objects, so that they can be transported over networks at appropriate quality of service levels.

The visual standard will include all the coding tools relating to visual data, natural and synthetic. Currently there are two verification model (VM) specifications — the video VM for natural video coding, frame based or arbitrarily shaped; and the synthetic/natural hybrid coding (SNHC) VM for synthetic 2D/3D graphics tools. As the tools are tested and proven within these VMs, they will be incorporated into the visual WD, and ultimately, if widely agreed on, in the IS.

For the highest efficiency in video compression the video VM is building on the work of ITU-T Recommendation H.263. Extensions have already been made to the core coder, most significantly the ability to code shape and transparency information of so-called video object planes or VOPs (see Fig 11). The video coder will be capable of coding video at rates from 10 kbit/s to more than 4 Mbit/s.

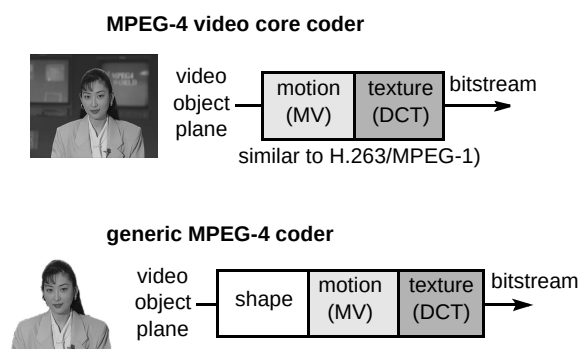


Fig 11 Video verification model stages.

Transcoding between MPEG-4 and MPEG-1/H.263 elementary bit streams should be straightforward in the frame-based (or rectangular VOP) operating mode (e.g. when using the proposed real-time communications profile). The addition of arbitrary shape coding in the generic coder enables a range of new content-based functionalities such as content manipulation in the compressed domain, and content scalability. The shape coding is performed using a bitmap-based technique known as context-based arithmetic coding.

The visual standard will not specify how shape and alpha (transparency information) planes are to be generated as this is a producer or encoder issue. MPEG philosophy is that only decoder issues should be specified to guarantee interworking — enabling competition between companies as to which can provide the best encoder engine, and also, in this instance, provide the most useful shape and alpha data.

widespread use in the television industry, are one source for this information. The video group is considering automatic video segmentation algorithms which may well form an informative annex to the standard to help content producers. Other automatic and semi-automatic segmentation techniques are also being investigated by the European COST211 ter [26].

The synthetic video components of the SNHC VM currently include media integration of text and graphics (MITG), face and body animation, texture coding (generic and view-dependent textures) and static and dynamic mesh coding with texture mapping. MITG provides the capability to overlay and scroll text, images and graphics on coded video backgrounds. Work on this is currently proceeding and is likely to be included soon in the visual WD.

Face animation will allow definition and animation of synthetic 'talking heads' as shown in Fig 12. Only the face definition parameters (FDPs) and the face animation parameters (FAPs) need standardising here. Work at BT Laboratories [25, 27] combining this with texture mapping (Fig 13), has shown how realistic these synthetic personae can actually be. Body animation issues have not been worked on yet but one proposal in this area known as Jack [28] has already been input to the group.

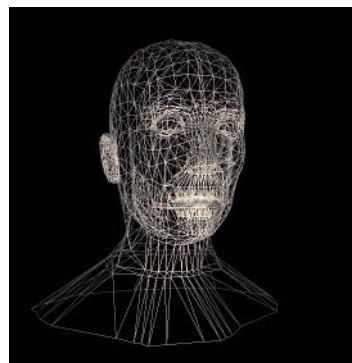


Fig 12 Wireframe head model.



Fig 13 Texture-mapped model.

5.2 MPEG-7

The title of the MPEG-7 activity is 'Multimedia Content Description Interface' [29]. The focus of this new work item is no longer that of efficient *compression* of audio/visual content, but rather its representation for *searching* and *browsing* purposes. This is a very real need with the rapid deployment of digital multimedia databases including sound and images on the World Wide Web (WWW), video-on-demand systems, and corporate image databases — hence the ambitious time-scale of this activity to just one year after ISO/MPEG-4 is finalised.

What MPEG-7 does inherit from previous MPEG activities is the requirement that only the minimum needed for interoperability is to be standardised. For example, it may specify particular sets of audio and video features that can be used as descriptors, such as shape outlines, colour histograms, frequency components, and how they are associated with the original content. However, it will not specify how to extract these video or audio features required for content description, in the same way that the analysis and encoding stages are not specified for the earlier compression-based standards. Nor will it specify a search engine or user interface — which might in practice take the form of a web browser or an intelligent agent. MPEG-7 will concern itself solely with a standardised description of audio/visual content features and will thereby be able to benefit from future progress in automatic feature extraction research. Both information extracted from the scene (e.g. colour, shape, texture) and external information about the scene (e.g. textual annotations, script) are expected to be represented, although the work is at a very early stage at the time of writing.

The standard will have many application domains but a major one is likely to be WWW search engine functionality on audio, video and still pictures. Nearly all WWW searching is currently done on a text basis, although image-based searching is starting to appear in a basic form [30]. MPEG-7 is currently in a requirements capture phase of development but is scheduled to reach 'International Standard' status in the year 2000.

6. Conclusions

Video coding has been one of the success stories of the last decade, enabling a host of applications including PC-based multimedia, videotelephony, highly flexible video-on-demand, and digital broadcasting, that were not possible with the previous analogue formats or uncompressed digital video. The need for compression comes from the limited bit rates of transmission channels, and the cost of digital storage. Although both of these will continue to increase in capacity and reduce in price, we still seem to be some way from the time when the cost will be so low that

compression is not needed. Even then, some means of delivery such as radio and satellite channels will continue to have restricted bandwidths.

The successful market development in many areas has been dependent upon agreed standards, principally from the ITU and ISO/MPEG, although particularly for PC-based applications, proprietary algorithms have also found a place.

Conventional compression algorithms seem to be reaching an asymptote, and although MPEG-4 will provide more functionality in terms of being able to define and code objects within a scene separately, in terms of absolute compression it will not provide very much gain over MPEG-2 or H.263. New techniques in MPEG-4 for Synthetic/Natural Hybrid Coding will push compression further, but only in limited domains such as human head/body modelling, and synthetically generated video. Nevertheless, video coding remains an active area of research with, for example, the full potential of wavelet-based schemes yet to be realised, and it is expected that compression ratios will continue to inch upwards.

Using real video in a teleconference has the distinct advantage that within the bounds of coding distortion there is no direct possibility of misrepresentation of the participant's appearance including facial expressions and body language, and brings a feeling of trust to the teleconference. A purely synthetic avatar in a virtual teleconference could be made to do anything, but the complete independence from one's surroundings and the physical world that an avatar brings makes new teleconferencing paradigms possible.

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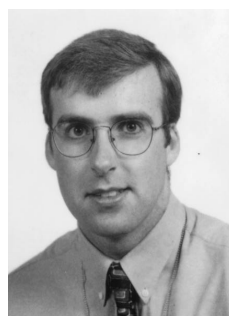
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Mike Whybray joined BT Laboratories in 1977, working on reliability physics, and in 1983 moved to the visual telecommunications division where he led a group whose activities included developing videophones for deaf people, writing software for DSP-based videophones, and building demonstrators of intelligent surveillance systems and synthetic 'talking head' displays. He currently leads a group developing new speech and image coding algorithms and standards. He is a Chartered Engineer and a Member of the IEE.



Geoff Morrison graduated from St John's College, Cambridge in 1971 and joined BT initially working on analogue techniques for video transmission and storage. From 1980 he headed a group which designed, constructed and installed ten analogue and 2-Mbit/s video switches together with the associated software control system for BT's visual services trial. In 1985 he spent 6 months with NTT Labs in Japan. Since then he has been heavily involved directly in the ITU-T and ISO standardisation bodies concerning digital video compression, and in various supporting European collaborative projects.



Patrick Mulroy received a 1st class Honours degree in Electrical and Electronic Engineering from Heriot-Watt University in 1992, and joined the image coding and processing group at BT Laboratories. Here he has worked on image compression algorithms including implementation of H.261 on a parallel TMS320C40 system, development and testing of H.263 (particularly arithmetic coding), and on object segmentation and tracking. He is the BT representative on the COST211 ter project, and MPEG-4 and an associate member of the IEE.